Şerban-Teodor Nicolescu

Engineer

Dedicated engineer leveraging multidisciplinary expertise and an extensive repertoire of skills to enhance performance and drive growth across a wide array of industries

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Profile / Summary

Medical engineer and PhD candidate specializing in Artificial Intelligence-Based Human Behaviour Assessment Systems, with a multidisciplinary background spanning research and development, information technology, and medical engineering. My professional experience encompasses physiological signal acquisition and processing, machine learning model development, IT system administration, and medical equipment service engineering. Throughout my career, I have developed strong analytical, technical, and problem-solving skills, complemented by practical experience in programming, software development, and multidisciplinary collaboration. My expertise in artificial intelligence, machine learning, and IT technologies enables me to approach complex technical challenges from multiple perspectives. Driven by continuous learning, adaptability, and professional development, I am committed to applying my knowledge and experience to deliver innovative, effective, and reliable solutions across diverse engineering and technology environments.

Key Skills / Competencies

- Problem-solving
- Critical Thinking
- Analytical Thinking
- Creative Thinking
- Communication
- Active Listening
- Adaptability
- Teamwork and collaboration
- Time management
- Perseverance
- Self-motivation
- Self-awareness
- Self-learning
- Patience
- Empathy

Education / Qualifications

Oct 2022 - Jul 2027 – University POLITEHNICA of Bucharest

PhD – Artificial Intelligence-Based Human Behaviour Assessment Systems

Dec 2024 - Mar 2026 – Devmind Academy

Junior Backend Developer: Core Java, OOP, Data Structures, Multithreading, SQL & JDBC

Full-Stack Developer: Spring Boot, REST APIs, JPA/Hibernate, React, Docker & CI/CD

Oct 2020 - Jul 2022 – University POLITEHNICA of Bucharest - Faculty of Medical Engineering

MEng – Modern Technologies for Medical Engineering Program

Oct 2016 - Jul 2020 – University POLITEHNICA of Bucharest - Faculty of Medical Engineering

BEng – Specialization in Medical Systems and Equipments

Career History / Work Experience

Feb 2017 - Present – ARC-ROM TRADING, Bucharest, Romania - System Administrator

ARC-ROM TRADING has established itself as a Leading Supplier in the Romanian market, providing Machinery, Equipment, and Consumables for the Extraction, Processing, Montage, Treatment, Cleaning, and Maintenance of Natural and Engineered Stones.

- Within the scope of my professional responsibilities, I undertake a range of essential tasks, including installing and configuring hardware and software, monitoring system performance, troubleshooting technical issues, implementing robust security measures, backing up critical data, managing user accounts and permissions, as well as planning and executing new system implementations and upgrades.

Nov 2021 – Jan 2024 – WING Group, Bucharest, Romania - Medical Engineer

With almost two decades of experience and a portfolio of 70 significant projects worth €192 million, WING is the Leading Research, Development, and Innovation Group in Romania.

- I am part of the interdisciplinary research team focused on the development of an Intelligent System for Automatic Assessment and Interpretation of Disguised Behaviour for Critical Infrastructure Personnel.
- My main responsibilities as part of the R&D team include:
  - Acquisition, preprocessing, postprocessing and analysis of multimodal physiological signals
  - Building and augmenting physiological signals datasets for machine learning
  - Developing, training and optimizing ML models for multimodal physiological signal classification and emotion recognition

Jun 2019 - Aug 2019 – GE HealthCare, Bucharest, Romania - Internship - Field Service Engineer

General Electric Healthcare is a division of General Electric (GE) that specializes in providing Medical Imaging and Diagnostic Equipment, Patient Monitoring Systems, Healthcare IT Solutions and Services, as well as other Healthcare Technologies.

- Assisted the field service engineer teams to perform installation, troubleshooting, maintenance, repair activities on designated medical imaging equipment: MRI, CT, SPECT-CT, X-Ray scanners and ICU ventilation systems.

IT Skills / Software

- Java
- Python
- PyTorch
- Artificial Intelligence
- Machine Learning
- Deep Learning
- LabVIEW
- MATLAB
- Simulink

Languages

- English: Full Professional Proficiency

Additional Education / Qualifications

Apr 2023 – Athena - Training and Development Center

Module Trainer Diploma, accredited by the N.C.A (National Council for Adult Training), and implicitly recognized by the Ministries of Labor and Education.

Mar 2021 - Apr 2021 – SofMedica - ISLE Academy

Attended courses on finance, sales, marketing in the medical equipment field, theoretical presentations, practical demonstrations and hands-on sessions with state of the art medical devices such as the Da Vinci surgical system.

Oct 2017 - Nov 2017 – University POLITEHNICA of Bucharest - Faculty of Industrial Engineering and Robotics

Autodesk Certified Inventor Training Module